

SWETHA PANDALA

Full Stack Software Engineer | Generative AI Developer

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PROFESSIONAL SUMMARY

- Full Stack Software Engineer and Generative AI Developer with around 7 years of experience designing, building, and deploying scalable enterprise applications across the full stack, cloud platforms, and AI-native environments, with strong exposure to financial services and healthcare domains.
- Strong backend engineer proficient in Java 8/11/17, Spring Boot, Spring Cloud, and microservices architecture, complemented by deep Python expertise using FastAPI and Flask to deliver high-performance, production-grade REST APIs and data services.
- Experienced front-end developer skilled in React.js, JavaScript (ES6+), and TypeScript, building responsive dashboards, reusable component libraries, and data visualization interfaces for operational and analytical reporting.
- Hands-on Generative AI engineer building autonomous and multi-agent systems with orchestration using AWS Bedrock AgentCore, Strands Agents, LangChain, and LangGraph, implementing tool use, shared state, and multi-step reasoning.
- Built production Retrieval-Augmented Generation (RAG) pipelines and knowledge bases using Amazon Bedrock Knowledge Bases and OpenSearch Serverless as a vector database to deliver grounded, citation-backed, context-aware responses.
- Implemented agent observability, tracing, and automated evaluation (evals) using LangSmith to monitor accuracy, latency, and cost across LLM and agentic workflows.
- Skilled in leveraging AI coding tools including GitHub Copilot (agent mode), Claude (Anthropic), and Cursor to build coding agents, accelerate feature delivery, and automate code review and debugging.
- Proven experience deploying cloud-native applications on AWS (EC2, RDS, Lambda, S3, ECS/EKS, Bedrock, OpenSearch Serverless) and Azure with full CI/CD automation via Jenkins, GitLab CI, and GitHub Actions.
- Solid expertise across relational and NoSQL databases including PostgreSQL, MySQL, MongoDB, DynamoDB, and Cassandra, with strong data modeling, indexing, and query optimization skills.
- Experienced across the full software development lifecycle including analysis, design, development, testing, performance tuning, deployment, and production support of high-volume enterprise applications in Agile/Scrum environments.

TECHNICAL SKILLS

Languages: Java 8/11/17, Python 3.x, JavaScript (ES6+), TypeScript, SQL, R

Backend & APIs: Spring Boot, Spring MVC, Spring Cloud, Spring Security, Microservices, FastAPI, Flask, Node.js, REST APIs, GraphQL, SOAP

Frontend: React.js, Angular, HTML5, CSS3, Bootstrap, Material UI, Redux, AJAX, JSON

Generative AI & Agents: AWS Bedrock AgentCore, Strands Agents, LangChain, LangGraph, LangSmith, Multi-Agent Orchestration, Autonomous Agents, RAG Pipelines, Prompt Engineering, LLM Evals, Amazon Bedrock Knowledge Bases, Model Context Protocol (MCP)

LLMs & Embeddings: Amazon Bedrock (Claude, Titan, Llama), OpenAI GPT-4, Anthropic Claude, Hugging Face Transformers, Sentence Transformers

Vector & Search: OpenSearch Serverless (vector store), Pinecone, pgvector, FAISS, Amazon OpenSearch Service, semantic search

AI Coding Tools: GitHub Copilot (agent mode), Claude (Anthropic), Cursor, Tabnine, Codeium, AI-native development workflows

ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, PySpark

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Cassandra, Oracle

Cloud Platforms: AWS (EC2, S3, RDS, Lambda, ECS/EKS, Bedrock, OpenSearch Serverless, SQS, CloudWatch, IAM), Microsoft Azure

DevOps & CI/CD: Jenkins, GitLab CI, GitHub Actions, Docker, Kubernetes, Terraform, Ansible, CloudFormation

Messaging & Streaming: Apache Kafka, AWS SQS

Version Control: Git, GitHub, GitLab, Bitbucket, GitFlow branching strategy

Monitoring & Logging: LangSmith, New Relic, Splunk, CloudWatch, ELK Stack, Prometheus

Project & Process: Agile/Scrum, Jira, Confluence, Trello, Slack, SDLC, TDD

PROFESSIONAL EXPERIENCE

Software Engineer | MetLife, Whippany, New Jersey

Nov 2024 – Present

- Designed and orchestrated multi-agent systems composed of planner, retriever, and executor agents that coordinate through shared state and message passing, enabling complex task decomposition and multi-step reasoning.
- Engineered stateful orchestration layers with LangChain and LangGraph to model cyclic agent workflows featuring conditional routing, human-in-the-loop checkpoints, and durable, resumable execution.
- Developed Retrieval-Augmented Generation (RAG) pipelines combining Amazon Bedrock Knowledge Bases with OpenSearch Serverless as the vector database, delivering grounded, citation-backed answers over proprietary financial documents.
- Created and managed Amazon Bedrock Knowledge Bases with automated ingestion, chunking, and embedding pipelines, provisioning OpenSearch Serverless vector collections for low-latency semantic search and retrieval.
- Implemented end-to-end observability, tracing, and debugging of agent and LLM workflows using LangSmith, capturing token usage, latency, tool calls, and failure modes across distributed agent runs.
- Built automated evaluation (evals) frameworks using LangSmith datasets and LLM-as-judge scoring to measure answer relevance, faithfulness, and regression across prompt and model versions before promotion to production.
- Developed coding agents and automation using GitHub Copilot agent mode to scaffold services, generate unit tests, and execute multi-file changes, accelerating delivery and standardizing patterns across the engineering team.
- Exposed agent and GenAI capabilities through Python FastAPI microservices integrated with Bedrock foundation models (Claude, Titan), implementing streaming responses, strict schema validation, and guardrails for safe outputs.
- Integrated GenAI services into existing enterprise systems by building Java Spring Boot microservices that consume agent endpoints via REST APIs and event-driven messaging, balancing full-stack and AI responsibilities.
- Applied prompt engineering, reusable prompt templates, and structured output parsing to ensure deterministic, schema-compliant responses from foundation models in regulated financial use cases.
- Collaborated with data science, risk, and compliance teams to align agentic AI solutions with model governance, auditability, and responsible-AI standards across the mortgage technology platform.
- Designed and implemented RESTful APIs in Java and Spring Boot to enable seamless, secure data exchange between distributed systems and applications, ensuring efficient communication and integration.
- Designed and developed scalable backend services and automation workflows using Python 3.x, FastAPI, and Flask, delivering production-grade REST APIs integrated with financial and healthcare data platforms.
- Architected and delivered production-grade Python microservices with FastAPI and Flask, designing modular, reusable service components that reduced backend development time by 35% across the engineering team.

- Built and enhanced front-end applications using React.js and JavaScript, creating reusable component libraries, dynamic dashboards, and data visualization interfaces for operational and financial reporting.
- Integrated GitHub Copilot, Claude and Cursor into the daily development workflow to accelerate feature delivery, automate boilerplate generation, and improve code review quality, reducing implementation time by 30%.
- Built and maintained RAG pipelines combining vector databases (Pinecone, pgvector) and LLMs (GPT-4, Claude) to deliver context-aware, domain-specific responses grounded in proprietary financial and healthcare data.
- Developed AI-assisted features including intelligent search, document summarization, and workflow automation using LLM APIs and prompt engineering best practices.
- Built high-performance Python data processing pipelines handling 2M+ financial transaction records daily, implementing async processing, connection pooling, and batch optimization for low-latency throughput.
- Developed Python REST and GraphQL APIs consumed by internal dashboards, third-party fintech integrations, and React.js frontends, enforcing strict schema validation, versioning, and error-handling standards.
- Designed and deployed applications on AWS using EC2, RDS (PostgreSQL/MySQL), S3, Lambda, ECS, and CloudWatch, implementing auto-scaling, IAM policies, and high-availability architectures.

Software Engineer | Morgan Stanley, New York, New York

Aug 2022 – Oct 2024

- Built scalable microservices using Java 8–17, Spring Boot, Spring MVC, Spring Security, and Spring Cloud, applying strong RESTful API design, data modeling, and performance-optimization practices.
- Refactored legacy Java monolith modules into Python and Spring Boot microservices, improving maintainability, reducing deployment complexity, and cutting average service startup time by 60%.
- Implemented Python Celery task queues with a Redis broker for asynchronous background job processing, handling scheduled financial reports, notification workflows, and retry logic for failed transactions.
- Implemented and managed CI/CD pipelines using Jenkins and GitHub Actions, automating build, test, and deployment stages across dev, staging, and production, cutting release cycle time by 45%.
- Worked with PostgreSQL and MongoDB for financial data storage, designing optimized schemas, complex queries, and data migration scripts for transactions, events, and audit logs.
- Developed autonomous AI agents with tool usage, context retention, and multi-step reasoning to streamline internal document workflows, data extraction, and intelligent task automation.
- Applied prompt engineering techniques to design context-aware prompts for code generation, test-case generation, API workflows, and data-processing use cases using Claude and OpenAI APIs.
- Containerized Python and Java microservices using Docker and deployed to Kubernetes/EKS clusters, improving release consistency and supporting blue-green deployment strategies.
- Built event-driven data pipelines using Apache Kafka for real-time event streaming, supporting monitoring, alert workflows, and audit-trail management across distributed services.

Java/Python Full Stack Developer | Optum | India

Sep 2020 – Jun 2021

- Developed full stack web applications using Java, Spring Boot, Python (Django/Flask), and React.js, delivering RESTful APIs and dynamic UIs for data-intensive banking and healthcare analytics platforms.
- Built scalable Spring Boot microservices and Python services, applying clean architecture, dependency injection, and layered design for maintainable, testable enterprise code.
- Built and optimized database layers using PostgreSQL and MongoDB, designing efficient data models, indexing strategies, and stored procedures that reduced query response times by 40%.
- Implemented OAuth 2.0 and JWT-based authentication for internal and external REST APIs, ensuring secure, role-based access across enterprise banking and healthcare integrations.

- Deployed applications on AWS using Elastic Beanstalk, RDS, S3, and CloudWatch, configuring auto-scaling groups and health monitoring for production environments.
- Built CI/CD automation using Jenkins and GitHub Actions, improving deployment reliability and reducing manual release effort by 50% through automated test execution and environment promotion.
- Integrated Redis as a distributed caching layer to improve API response times by 40% and reduce database load on high-frequency data access patterns.
- Applied scikit-learn and Python-based ML pipelines for analytics use cases including risk scoring and data classification with 85%+ model accuracy.

Web Application Developer | ValueMomentum | India

Feb 2020 – Aug 2020

- Developed reusable UI components using React.js, Angular, HTML5, CSS3, and JavaScript, improving maintainability and consistency across enterprise web applications.
- Built backend Python and Java services for CRUD operations, business workflows, and data processing, integrated with MySQL and MongoDB databases.
- Optimized SQL queries, joins, and indexes across MySQL and PostgreSQL databases, reducing response times by 20–30% for reporting modules.
- Assisted in deploying applications to AWS cloud environments and supported CI/CD pipeline setup using Jenkins for automated builds and deployments.
- Collaborated with QA and cross-functional Agile teams to validate requirements, troubleshoot issues, and deliver application features within sprint timelines.

Java Developer | Abbott | India

Nov 2018 – Jan 2020

- Built REST microservices in Java and Python integrated with MySQL, MongoDB, and Cassandra for high-volume transactional and clinical data workflows.
- Designed and implemented RESTful APIs using Spring Boot and Python Flask to expose data services consumed by internal analytics dashboards and external client portals.
- Developed Python-based data validation, transformation, and automation scripts, improving processing accuracy and reducing manual review effort by 30%.
- Built data ingestion pipelines using Python and PySpark to process 1M+ records, transforming raw data into structured formats for analytics and compliance reporting.
- Applied scikit-learn models for risk classification, achieving 82% accuracy on structured datasets used for population-health and reporting use cases.
- Integrated applications with AWS services including S3 for data storage and RDS (MySQL) for transactional data, supporting scalable and secure cloud-based workflows.
- Implemented logging, monitoring, and alerting using AWS CloudWatch and the ELK Stack to maintain production stability and support SLA-driven incident resolution.
- Used Jenkins CI/CD pipelines to automate build and deployment of microservices, reducing manual deployment errors and cutting release cycle time by 35%.
- Participated in all SDLC phases including requirements gathering, design, development, UAT support, and production deployment within Agile/Scrum teams using Git, Bitbucket, and Jira.

KEY PROJECTS

Enterprise Multi-Agent Financial Assistant (AWS Bedrock AgentCore)

- Built a multi-agent assistant on Amazon Bedrock AgentCore using Strands Agents and LangGraph orchestration, with specialized agents for retrieval, analysis, and report generation over financial documents.
- Grounded responses with a RAG pipeline backed by Amazon Bedrock Knowledge Bases and OpenSearch Serverless vector collections, and monitored quality and cost with LangSmith tracing and automated evals.

Document Intelligence Platform (RAG)

- Built a Python RAG pipeline using LangChain, GPT-4, and pgvector (PostgreSQL) to enable semantic search and intelligent Q&A over 500K+ documents and records.
- Deployed on AWS (EC2, S3, RDS) with a React.js frontend, delivering context-aware responses grounded in proprietary data with secure, compliant data handling throughout.

- Integrated Claude and GitHub Copilot across the development lifecycle, reducing implementation time by 35% and improving code quality through AI-assisted reviews.

Swift Transaction Prediction Engine (GPT-2 Fine-Tuning)

- Built a Python transaction-status prediction system using a fine-tuned GPT-2 model (Hugging Face Transformers) trained on historical financial transaction data.
- Implemented NLP-based query handling with FuzzyWuzzy and transformer tokenization; managed data in Apache Cassandra with Matplotlib-based training visualization.

Quora Question Similarity — NLP Classification

- Developed a Python logistic regression model to determine semantic similarity of 400K+ question pairs, achieving a log-loss score of 0.284 using TF-IDF weighted GloVe embeddings.

EDUCATION

Doctorate in Business Administration with Applied Artificial Intelligence

Belhaven University, Jackson, MS

Expected 2029

Master's in Information Systems

University of Memphis, Memphis, TN

GPA: 3.8/4.0

Dec 2022

Bachelors in Computer Science and Engineering

JNTUH, Hyderabad, India

GPA: 9.0/10

Aug 2020

CERTIFICATIONS

- AWS Certified Developer – Associate (2024)
- Java Programming Certificate (2021)
- HTML, CSS, and JavaScript for Web Developers (2020)
- Python Programming Certificate (2020)

ACTIVITIES & PROFESSIONAL DEVELOPMENT

- Attended Artificial Intelligence and Machine Learning workshops conducted by Internshala (2019).